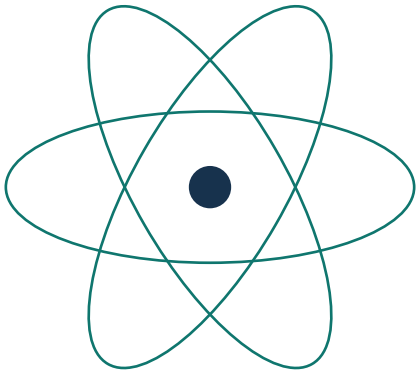
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



NSMO Collection I

Olympiad Problem Collection

Olympiad Training • Mechanics and Fluids

SPhOH Team

FIRST EDITION • 2026

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About This Collection

This collection is part of the training resources developed by the **Saudi Physics Olympiad Hub (SPhOH)**. It is designed to support students preparing for the **first stage of the National Science and Mathematics Olympiad (NSMO)**, with a particular focus on the physics-related mathematical tools and fundamental concepts that underpin problem-solving.

The collection covers four areas: **vectors, kinematics, unit conversions, and fluid mechanics**. Together, these topics develop essential skills in mathematical representation, physical reasoning, quantitative analysis, and the application of fundamental principles to unfamiliar problems.

The collection is **problem-centered**. Rather than presenting an extensive textbook-style treatment, it emphasizes active problem-solving, with concise notes provided to introduce key concepts and clarify useful techniques. Hints are included to help students overcome specific difficulties while encouraging independent reasoning.

This collection is intended as a supplementary training resource, not a replacement for the official NSMO syllabus or other preparation materials.

How to Use This Collection

Attempt each problem independently before consulting its hint. Begin by identifying the relevant physical principles, defining the quantities involved, and choosing an appropriate mathematical approach.

If you become stuck, use the hint to identify a possible next step rather than reading it as a complete solution. After resolving the difficulty, return to the problem and complete the reasoning independently. Where appropriate, check the consistency of your result, including its units, sign, magnitude, and physical interpretation.

Difficulty Levels

The problems are organized into three training levels:

- **Level I** — Fundamental problems designed to reinforce essential concepts and mathematical techniques.
- **Level II** — Intermediate problems requiring multiple steps, connections between concepts, or more substantial reasoning.
- **Level III** — More challenging problems emphasizing deeper analysis, strategic problem-solving, and the application of familiar ideas in less familiar settings.

These levels indicate the relative difficulty of the problems within this collection; they do not represent official NSMO difficulty classifications.

For more visit: https://devfhd.github.io/SPhOH/

Part I

Units Conversion

CHAPTER 1

Units Conversion

Topic Note: The Chain-Link Method
Unit conversion is not merely shifting decimal points; it is the mathematical process of multiplying a quantity by a conversion factor equal to 1. Mastering the behavior of nested prefixes, squared/cubed units, and inverse proportionalities is a core survival skill in advanced physics competitions.

Reference: Essential Conversion Factors

Category	Unit / Quantity	Conversion Equivalent
Length	1 Kilometer (km)	$= 10^3$ m
	1 Meter (m)	$= 100$ cm $= 10^3$ mm $= 10^6 \mu$m
	1 Parsec (pc)	≈ 3.26 light-years (ly)
Volume	1 Cubic Meter (m^3)	$= 1000$ Liters (L)
	1 Liter (L)	$= 10^3$ cm^3 $= 10^{-3}$ m^3
	1 Fluid Ounce (fl oz)	≈ 29.57 cm^3
Time	1 Hour (hr)	$= 60$ min $= 3600$ s
Mass	1 Kilogram (kg)	$= 10^3$ g
	1 Gram (g)	$= 10^3$ mg
Pressure	1 Atmosphere (atm)	$= 101,325$ Pascals (Pa)

1.1 Multiple Choice Questions

PROBLEM 1

A

Level I

n engine's temperature increases by 10°C. What is this increase (change) in degrees Fahrenheit (°F)?

(a) 50°F

(b) 18°F

(c) −12°F

(d) 283°F

PROBLEM 2

T

Level II

he density of a certain material is 1 mg/mm^3. What is the value of this density in fundamental SI units (kg/m^3)?

(a) 1 kg/m^3

(b) 100 kg/m^3

(c) $1,000\ kg/m^3$

(d) $1,000,000\ kg/m^3$

PROBLEM 3

I

Level II

n some countries, fuel efficiency is measured in km/L, while in others it is measured in L/100km. If a car's efficiency is 20 km/L, what is its efficiency in the latter unit?

(a) 2000 L/100km

(b) 0.05 L/100km

(c) 5 L/100km

(d) 50 L/100km

PROBLEM 4

T

Level II

---

he acceleration due to gravity is approximately 10 m/s^2. If we want to express it in km/min^2, we must:

(a) Multiply by 3.6

(b) Divide by 1000, then multiply by 60

(c) Divide by 1000, then multiply by 3600

(d) Multiply by 1000, then divide by 3600

1.2 Free-Response Questions

PROBLEM 5

I

Level I

---

n a hospital, a patient is given an intravenous fluid at a flow rate of 2 fluid ounces per hour. Given that 1 fluid ounce $\approx 29.57\ cm^3$.
Required: Calculate the flow rate in cubic millimeters per second (mm^3/s).

PROBLEM 6

A

Level II

---

stronauts discover an alien civilization that uses a completely different system of physical units:

- The unit of mass is the "Zorp" (Z), where 1 Z = 1.5 kg.
- The unit of length is the "Blip" (B), where 1 B = 0.5 m.
- The unit of time is the "Glop" (G), where 1 G = 2 s.

A force of 10 Newtons ($\mathrm{N} = \mathrm{kg} \cdot \mathrm{m/s^2}$) acts on an object.
Required: Convert this force into the alien units ($\mathrm{Z} \cdot \mathrm{B/G^2}$), clearly showing your mathematical steps.

PROBLEM 7

I

Level II

---

n the van der Waals equation for real gases, the constant a for oxygen gas has the following value in SI units:

$$a = 0.138\ \mathrm{Pa} \cdot \mathrm{m^6/mol^2}$$

Given that atmospheric pressure is 1 atm = 101,325 Pa and 1 L = 10^{-3} $\mathrm{m^3}$.
Required: Convert this constant to the unit ($\mathrm{atm} \cdot \mathrm{L^2/mol^2}$).

PROBLEM 8

T

Level II

---

he radius of the observable universe is estimated to be 46.5×10^9 light-years (ly). Given that 1 Parsec (pc) = 3.26 ly.
Required: Calculate the volume of the observable universe (treating it as a sphere, $V = \frac{4}{3}\pi r^3$) in cubic parsecs ($\mathrm{pc^3}$).

Part II

Vectors

CHAPTER 2

Vectors

Topic Note.
Vectors represent quantities possessing both magnitude and direction. Master the decomposition into Cartesian components, operations in unit vector notation ($\hat{i}, \hat{j}, \hat{k}$), dot products (scalar projections and angles), and cross products (orthogonal vectors and geometric areas).

2.1 Vector Components

PROBLEM 9

LEVEL I

Angle from Vertical Axis

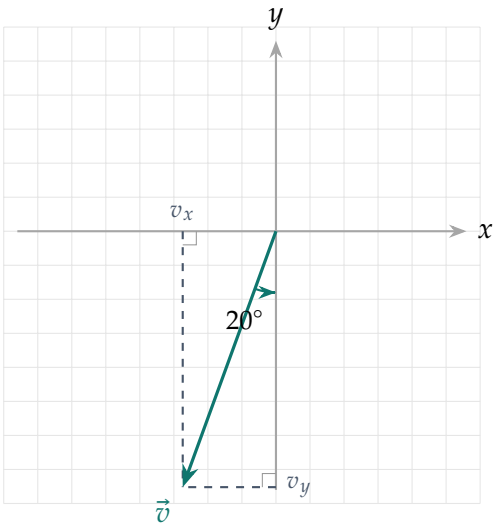
Find $\sin(\theta)$ (measured from the positive y-axis) for a vector whose x-component is 3 times larger than its y-component.

PROBLEM 10

LEVEL I

Polar Components and Sketch

What are the components of a vector with a magnitude of 10 units and a direction of 20° clockwise from the negative y-axis (−20° from the negative y-axis)? Sketch the vector on a 2D coordinate plane.



PROBLEM 11

LEVEL I

Position Vector Orientation

A position vector has components $d_x = -3$ m and $d_y = 4$ m.

(a) Find its magnitude.

(b) Find its direction relative to the positive x-axis.

(c) Find its direction relative to the negative y-axis.

PROBLEM 12

LEVEL I

Force Vector Decomposition

A force vector $\vec{F}$ has a magnitude of 25 N and points at an angle of 120° relative to the positive x-axis. Determine its scalar components F_x and F_y.

2.2 Unit Vectors

PROBLEM 13

LEVEL I

Unit Vector Notation of Force

A force vector with a magnitude of 10 N is directed at an angle $\theta = 45°$ from the positive x-axis. Express this force vector in terms of unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.

PROBLEM 14

LEVEL II

3D Velocity and Direction Angles

A velocity vector is given by $\vec{v} = (-5\hat{i} - 3\hat{j} + 2\hat{k})$ m/s. Find:

(a) The magnitude of the velocity vector.

(b) The direction angles between the vector and all three coordinate axes (x, y, and z).

(c) Sketch the vector along with its component projections on a 3D xyz-coordinate system.

PROBLEM 15

LEVEL II

Acceleration Vector from Component Ratios

Express an acceleration vector with a magnitude of 10 m/s^2 that satisfies the component relationship $a_y = 2a_z = 3.5a_x$. Furthermore, calculate the direction angles between the acceleration vector and all three coordinate axes.

PROBLEM 16

LEVEL III

Jerk Vector Representation

Jerk ($\vec{j}$) is defined as the time rate of change of acceleration. Express the jerk vector in unit vector notation under the following conditions:

(a) Motion with constant acceleration.

(b) The components satisfy $2j_y = 0.235j_z$, with $j_x = 0$.

(c) The magnitude of the jerk vector is 10 m/s^3, $j_y = 2$ m/s^3, and $\sin(\theta) = 3/5$, where θ is the angle between the vector and the xz-plane.

2.3 Vector Addition and Subtraction

PROBLEM 17

LEVEL I

Resultant of Two Vectors

Consider two displacement vectors $\vec{d}_1 = 3\hat{i} + \sqrt{55}\hat{j}$ and $\vec{d}_2 = -3\hat{i}$.

(a) Sketch the vectors on the xy-plane.

(b) Find the magnitude of the resultant vector $\vec{R} = \vec{d}_1 + \vec{d}_2$.

(c) Determine the direction angle of the resultant vector measured from the negative x-axis.

PROBLEM 18

LEVEL II

Vector Subtraction in 3D

Vector $\vec{C}$ is the resultant vector formed by adding vector $\vec{A}$ and vector $\vec{B}$ ($\vec{C} = \vec{A} + \vec{B}$). Given $\vec{C} = 5\hat{i} + 3\hat{k}$ and $\vec{A} = 2\hat{j} + 4\hat{k}$:

(a) Find vector $\vec{B}$ in unit vector notation.

(b) Calculate the magnitude and direction of vector $\vec{B}$.

PROBLEM 19

LEVEL II

Angle and Resultant Variation

Two vectors have magnitudes $A = 5$ m/s and $B = 10$ m/s.

(a) Find the angle θ between vectors $\vec{A}$ and $\vec{B}$ if the magnitude of their resultant $\vec{C}$ is 11 m/s.

(b) Find the resultant magnitude if $\theta = 90°$.

(c) Find the resultant magnitude if $\theta = -180°$.

(d) Find the resultant magnitude if $\theta = -45°$.

PROBLEM 20

LEVEL III

Algebraic Vector System and Projections

Given the vectors:

$$\vec{A} = x\hat{i} + (4y - 2)\hat{j} - 5\hat{k}$$

$$\vec{B} = (-2x)\hat{i} + 2\hat{j} + (1.5z + 4)\hat{k}$$

$$\vec{C} = \vec{A} - \vec{B} = (9 + x)\hat{i} + (7 - 2y)\hat{j} - 8\hat{k}$$

(a) Solve for the unknown variables x, y, and z.

(b) Draw vector $\vec{A}$ on the xy-plane.

(c) Draw vector $\vec{B}$ on the zy-plane.

(d) Draw the y-component of $\vec{C}$ on the xz-plane.

PROBLEM 21

LEVEL III

Three-Variable Vector Subtraction System

Given the vectors:

$$\vec{A} = 2a\hat{i} + 3b\hat{j} - c\hat{k}$$

$$\vec{B} = a\hat{i} - b\hat{j} + 2c\hat{k}$$

$$\vec{C} = 3\hat{i} + 2\hat{j} - 4\hat{k}$$

$$\vec{D} = \vec{A} - \vec{B} - \vec{C} = 5\hat{i} + 6\hat{j} + 3\hat{k}$$

(a) Determine the scalar constants a, b, and c.

(b) Express vectors $\vec{A}$ and $\vec{B}$ explicitly in unit vector notation.

(c) Calculate the magnitude of vector $\vec{D}$.

(d) Draw vector $\vec{C}$ on the xz-plane.

2.4 Dot Product

Topic Note: Scalar (Dot) Product
The dot product of two vectors $\vec{A}$ and $\vec{B}$ is a scalar quantity defined as:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos(\theta) = A_x B_x + A_y B_y + A_z B_z$$

Key applications include computing work $W = \vec{F} \cdot \vec{d}$, testing orthogonality ($\vec{A} \cdot \vec{B} = 0$), and finding projections.

PROBLEM 22

LEVEL I

Angle Between Two Vectors

Given two vectors $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = -2\hat{i} + 5\hat{j}$, calculate:

(a) The dot product $\vec{A} \cdot \vec{B}$.

(b) The angle θ between vectors $\vec{A}$ and $\vec{B}$.

PROBLEM 23

LEVEL I

Self-Dot Product Properties

What is the scalar result of forming the dot product of a general vector $\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$ with itself ($\vec{V} \cdot \vec{V}$)? Express your answer in terms of the magnitude $V = |\vec{V}|$.

PROBLEM 24

LEVEL II

Orthogonality Condition

Determine the value of the constant c such that the vectors $\vec{v}_1 = 2\hat{i} + c\hat{j} - 3\hat{k}$ and $\vec{v}_2 = c\hat{i} + 4\hat{j} + 2\hat{k}$ are perpendicular to each other.

PROBLEM 25

LEVEL II

Work Done by a Constant Force

A constant force $\vec{F} = (6\hat{i} - 2\hat{j} + 4\hat{k})$ N acts on an object, displacing it by $\vec{d} = (3\hat{i} + 5\hat{j} + 2\hat{k})$ m.

(a) Calculate the work done $W = \vec{F} \cdot \vec{d}$ by the force.

(b) Determine the angle θ between the force vector and the displacement vector.

PROBLEM 26

LEVEL III

Vector Projection

Given vectors $\vec{p} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{q} = 3\hat{i} + 4\hat{k}$:

(a) Find the scalar projection of $\vec{p}$ onto $\vec{q}$.

(b) Find the vector projection $\vec{q}$ onto $\vec{p}$.

(c) compare the results of a and b.

2.5 Cross Product

Topic Note: Vector (Cross) Product
The cross product of two vectors $\vec{A}$ and $\vec{B}$ yields a vector perpendicular to both, defined by the determinant:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Its magnitude is $|\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}| \sin(\theta)$, which equals the area of the parallelogram formed by $\vec{A}$ and $\vec{B}$.

PROBLEM 27

LEVEL I

Cross Product Calculation

Compute the cross product $\vec{A} \times \vec{B}$ for the vectors $\vec{A} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{B} = 4\hat{i} + 2\hat{j} - \hat{k}$.

PROBLEM 28

LEVEL I

Self-Cross Product Properties

What is the vector result of taking the cross product of any arbitrary vector $\vec{V}$ with itself $(\vec{V} \times \vec{V})$? Explain using both the magnitude equation and component definitions.

PROBLEM 29

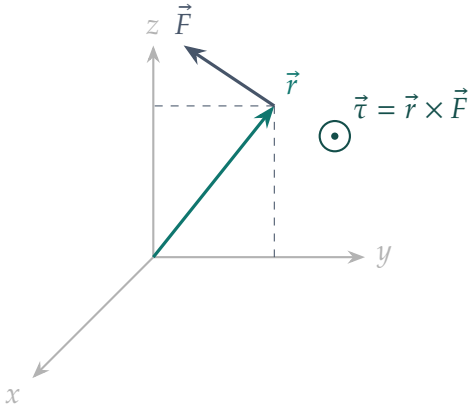
LEVEL II

Torque

Torque is a vector defined to be the cross product of moment arm and force, which can be written as: $\vec{\tau} = \vec{r} \times \vec{F}$. Consider that $\vec{r} = 4\hat{i} + 3\hat{j} + 5.5\hat{k}\ m$, and $\vec{F} = -6\hat{i} + 12\hat{j}\ N$. Find the torque in:

(a) unit vectors.

(b) direction and magnitude.



Find the area of the parallelogram defined by the vectors $\vec{a} = 3\hat{i} + 2\hat{j}$ and $\vec{b} = -\hat{i} + 4\hat{j}$.

PROBLEM 30

LEVEL III

Unit Normal Vector to a Plane

Find a unit vector perpendicular to both $\vec{u} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{v} = \hat{i} + 3\hat{j} + \hat{k}$.

Part III

Kinematics

CHAPTER 3

Kinematics

Topic Note: Kinematic Relations
Kinematics describes motion without considering the forces causing it. Key relationships include:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \qquad a = \frac{\Delta v}{\Delta t}.$$

For uniform acceleration a, where v_0 and v represent initial and final velocities, and x_0 and Δx represent initial position and displacement:

$$v = v_0 + at, \qquad x = x_0 + v_0 t + \frac{1}{2}at^2, \qquad v^2 = v_0^2 + 2a\Delta x.$$

In two-dimensional projectile motion, horizontal and vertical motions are independent:

$$x(t) = (v_0 \cos\theta)t, \qquad y(t) = (v_0 \sin\theta)t - \frac{1}{2}gt^2.$$

For a projectile launched and landing at the same height:

$$R = \frac{v_0^2 \sin(2\theta)}{g}, \qquad H = \frac{v_0^2 \sin^2\theta}{2g}.$$

3.1 1D Kinematics and Kinematic Equations

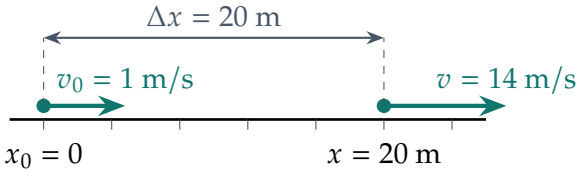
PROBLEM 31

LEVEL I

Acceleration along a Straight Line

Mohammed moves along a straight line with an initial velocity of 1 m/s. His velocity increases uniformly to 14 m while covering a distance of 20 m.

Determine the magnitude and direction of his acceleration.



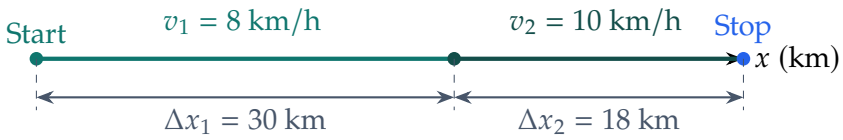
PROBLEM 32

LEVEL II

Multi-Stage Average Velocity

Hadi begins moving along a straight line at a constant speed of 8 1/h until he covers a distance of 30 . Immediately following this, he continues moving along the same line at a constant speed of 10 1/h for a distance of 18 , after which he comes to a complete stop.

Calculate the average speed for the entire trip.



PROBLEM 33

LEVEL II

Average Velocity over Split Distance and Time

A particle covers the first half of its total displacement with a constant velocity v_0. The second half of the distance is covered in two equal time intervals: moving with a constant velocity v_1 for the first half of that time, and with velocity v_2 for the remaining half of the time.

Calculate the average velocity v_{avg} over the entire duration of its motion.

PROBLEM 34

LEVEL II

Motorboat and Floating Raft

A motorboat traveling downstream along a river passes a floating raft at point A. The boat continues downstream for a time interval $\tau = 60$ min, then turns around and travels upstream at the same power setting, with constant speed relative to the water. The boat meets the raft again at a distance $l = 6.0$ downstream from point A.

Assuming the river flow velocity remains uniform and constant, determine the speed of the river current.

PROBLEM 35

LEVEL II

2D Motion under Constant Acceleration

A particle begins moving from an initial position vector

$$\vec{r}_0 = (3\hat{i} - 7\hat{j})\ \text{m}$$

with an initial velocity

$$\vec{v}_0 = 2\hat{i}\ \text{m/s}.$$

It moves with a constant acceleration

$$\vec{a} = (\hat{i} + 3\hat{j})\ \text{m/s}^2.$$

(a) Will the particle cross the x-axis? If so, at what time t and at what x-coordinate?

(b) Will the particle cross the y-axis? If so, at what time t and at what y-coordinate?

3.2 Vector Displacement in 3D Space

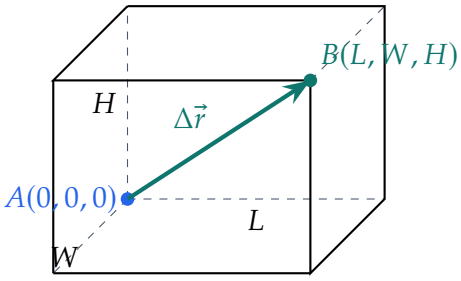
PROBLEM 36

LEVEL II

3D Box Corner-to-Corner Fly Displacement

A fly flies inside a rectangular room of dimensions $L \times W \times H$ from corner A, at the origin $(0, 0, 0)$, to the diagonally opposite corner B, at (L, W, H).

Find the magnitude of the fly's displacement vector $\Delta\vec{r}$.



3.3 Graphical Analysis of Motion

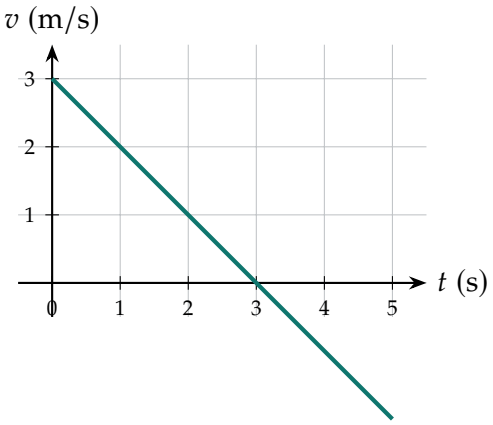
PROBLEM 37

LEVEL I

Position from Velocity–Time Graph

The graph below represents the velocity $v(t)$ of an object moving along a straight line as a function of time t. It is known that the position of the object at the instant when $v = 0$ is $x = 4$ m.

Determine the initial position of the object.



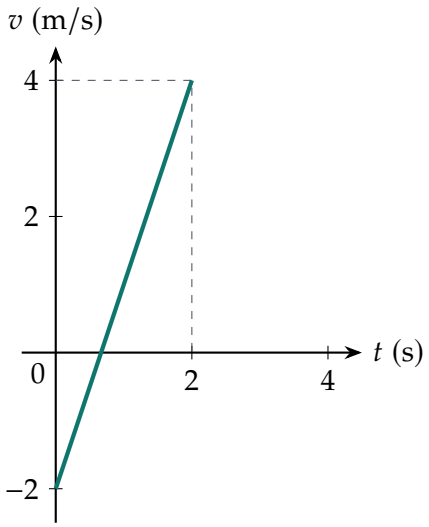
PROBLEM 38

LEVEL I

Acceleration from Velocity–Time Graph

The graph below depicts the velocity $v(t)$ of a particle moving along the x-axis over time.

Determine the instantaneous acceleration a at $t = 0$.

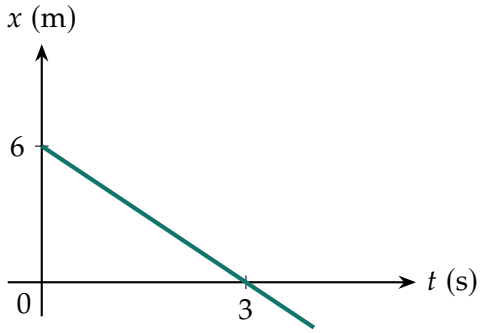


PROBLEM 39

LEVEL I

Instantaneous Velocity from Position–Time Graph

The position $x(t)$ of a particle moving along a straight line is plotted as a function of time t. Determine the instantaneous velocity at $t = 0$.



3.4 Free Fall and Vertical Motion under Gravity

PROBLEM 40

LEVEL III

Continuous Drop of Falling Balls

(a) A ball is released in air from rest. After time t_0, it falls a distance h. How much displacement does it cover during the second interval of duration t_0? Assume it has not reached the ground.

(b) Another ball is released from the ground with some initial speed. After time t_0, its height is h. What is its height after the second interval of duration t_0? Assume its velocity has not reached zero.

(c) A child standing on the roof of a residential building drops tennis balls vertically downward at regular time intervals. The rate is adjusted so that, at the exact instant a new ball is released, the previous ball is at an intermediate position in the air, while the ball before that has just struck the ground.
At the instant a ball is released, what is the height above the ground of the ball currently in the air? The building height is $H = 12$ m.

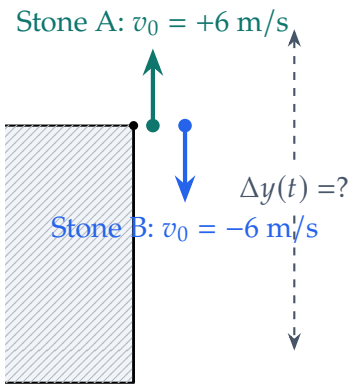
PROBLEM 41

LEVEL II

Separation of Oppositely Thrown Stones

A man standing at the edge of a tall vertical cliff throws two identical stones with the same initial speed $v_0 = 6$ m/s. Stone A is thrown vertically upward, while Stone B is thrown vertically downward.

Assuming neither stone has hit the ground, determine the distance between them after $t = 3$ s. Take $g = 9.8$ m/s^2.



PROBLEM 42

LEVEL II

Collision of Two Counter-Moving Balls

Two balls are thrown simultaneously along the same vertical line. Ball 1 is launched from the ground vertically upward with initial speed $v_1 = 20$ m/s. Ball 2 is released from a height $H = 150$ m, thrown vertically downward with initial speed $v_2 = 30$ m/s.

Take $g = 10$ m/s^2.

(a) Calculate the time t_{coll} until collision and the collision height y_{coll} above the ground.

(b) Which event occurs first: Ball 1 reaching its maximum height or the collision?

(c) Determine the distance travelled by Ball 1.

3.5 Projectile Motion

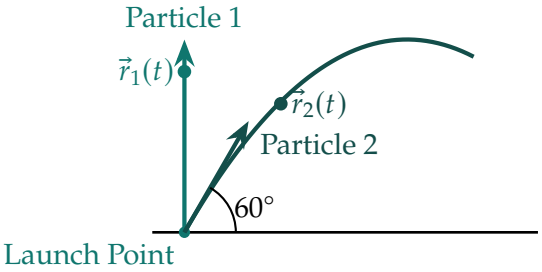
PROBLEM 43

LEVEL II

Diverging Projectiles Distance

Two particles are launched simultaneously from the same location with identical initial speed $v_0 = 25$ m/s. The first particle is projected vertically upward ($\theta_1 = 90°$), while the second is launched at $\theta_2 = 60°$ above the horizontal.

Ignoring air resistance, calculate the distance between the particles at $t = 1.70$ s.



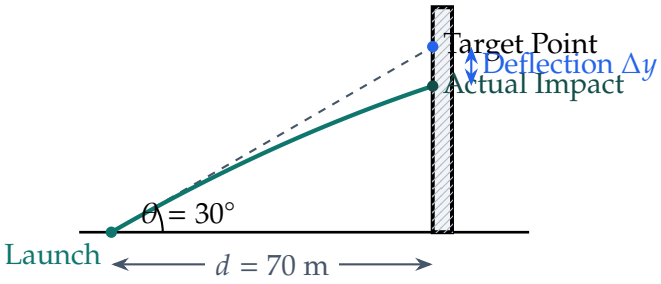
PROBLEM 44

LEVEL II

Direct Aim at a Vertical Wall

A projectile is launched from ground level at an angle $\theta = 30°$ above the horizontal with initial speed $v_0 = 40$ m/s, aimed directly at a target on a vertical wall a horizontal distance $d = 70$ m away.

Neglecting air resistance, calculate the vertical deflection from the target.



PROBLEM 45

LEVEL III

Cliff-Top Projectiles at Opposite Angles

Two identical balls are launched from the edge of a cliff at height $h = 40$ m above level ground with the same initial speed $v_0 = 30$ m/s. Ball 1 is launched at $\theta = 30°$ above the horizontal, while Ball 2 is launched at $\theta = 30°$ below the horizontal.

(a) Find the difference in flight times $\Delta t = t_1 - t_2$.

(b) Find the horizontal distance between their impact points.

(c) Compare the results with those for a projectile launched from level ground at 30° with the same initial speed. What do you observe?

PROBLEM 46

LEVEL II

Maximum Height to Range Ratio

A projectile is launched from level ground at an angle θ above the horizontal. Let R be its horizontal range and H its maximum height.

(a) Derive an expression for H/R in terms of θ.

(b) If $R = 2H$, find the launch angle θ.

PROBLEM 47

LEVEL II

Fixed Initial Speed with Variable Launch Angle

A cannon fires projectiles at fixed initial speed v_0, but the launch angle θ can be varied. The maximum possible horizontal range is L.

Determine the horizontal range of a projectile fired at $\theta = 30°$ in terms of L.

PROBLEM 48

LEVEL II

Level-Ground Projectile with Range 60 m

A projectile is launched from level ground at an initial angle $\theta = 45°$ and achieves a horizontal range $R = 60$ m. Neglect air resistance.

(a) Determine the initial speed v_0.

(b) Determine the total flight time T.

(c) How would the answers to (a) and (b) change if the range remained 60 m, but the launch angle increased to $\theta > 45°$? Explain physically.

PROBLEM 49

LEVEL II

Impact Velocity and Vector Direction

A particle is launched from level ground with initial velocity $\vec{v}_i$ at angle θ_i above the positive x-axis. When it returns to ground level, its final velocity is $\vec{v}_f$.

Express your answers in terms of v_i and θ_i.

(a) Find the angle between $\vec{v}_i$ and $\vec{v}_f$.

(b) Express $\vec{v}_i + \vec{v}_f$ using unit vectors $\hat{i}$ and $\hat{j}$.

PROBLEM 50

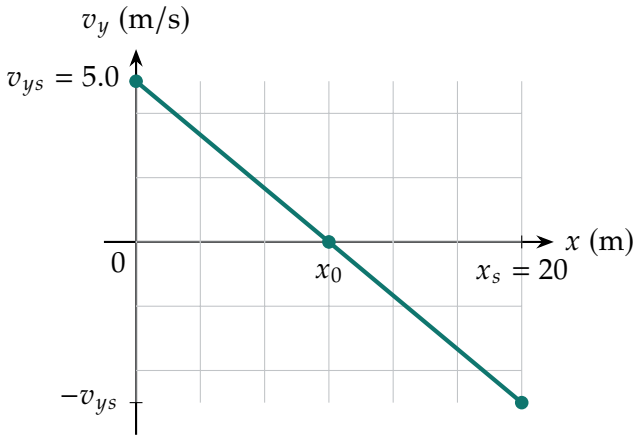
LEVEL III

Vertical Component of Impact Velocity against a Wall

A ball is shot from level ground toward a vertical wall located at distance x from the launch point. The vertical velocity component v_y just as the ball reaches the wall is plotted against x.

The graph scaling is $v_{ys} = 5.0$ m/s and $x_s = 20$ m.

Determine the launch angle θ_0.



Part IV

Fluid Mechanics and Statics

CHAPTER 4

Fluid Mechanics and Statics

4.1 Units and Density Conversion

Topic Note: Units and Density Conversion
Density is mass per unit volume:

$$\rho = \frac{m}{V}.$$

A common conversion is

$$1\ \text{g/cm}^3 = \frac{10^{-3}\ \text{kg}}{(10^{-2}\ \text{m})^3} = 1000\ \text{kg/m}^3.$$

To convert from kg/m^3 to g/cm^3, divide by 1000.

PROBLEM 51

LEVEL I

Density Unit Conversion I

An object has a density of 200 kg/m^3.
What is its equivalent density in g/cm^3?

PROBLEM 52

LEVEL II

Density Unit Conversion II

Oil has a density of 600 kg/m^3. Define the unit R by

$$1R = 600\ \text{kg/m}^3.$$

Express the density of water, 1000 kg/m^3, in terms of R.

4.2 Hydrostatic Pressure and Pascal's Principle

Topic Note: Pressure and Pascal's Principle
Pressure is force per unit area:

$$P = \frac{F}{A}.$$

For a static fluid of uniform density ρ, the pressure difference between two levels separated vertically by Δh is

$$\Delta P = \rho g \Delta h, \qquad P = P_0 + \rho g h.$$

Pascal's principle states that an applied pressure change is transmitted undiminished through an enclosed fluid:

$$\frac{f}{a} = \frac{F}{A} \quad \Longrightarrow \quad F = f\frac{A}{a}.$$

PROBLEM 53

LEVEL I

Pressure Under a Heavy Metallic Cylinder

A uniform metallic cylinder of mass 80 kg and height 2 m rests vertically on a planar base of area 25 cm^2.

Find the pressure exerted on the ground. Take $g = 9.8\ \text{m/s}^2$.

PROBLEM 54

LEVEL I

Force Scaling in a Hydraulic Press

A piston of cross-sectional area a exerts a force of magnitude f on the enclosed liquid. The liquid supports a larger piston of area A.

What force magnitude F can the larger piston sustain in static equilibrium? Neglect differences in height and piston weight.

PROBLEM 55

LEVEL II

Circular Hydraulic Lift

A hydraulic lift has two circular pistons. The smaller piston has diameter 5.0 cm, and the

larger piston has diameter 25.0 cm.

What force must be applied to the smaller piston to lift a 1500 kg vehicle? Take $g = 10\ \text{m/s}^2$.

Neglect piston weights and height differences.

PROBLEM 56

LEVEL II

Hydrostatic Pressure Difference and Depth

A vessel contains static fresh water of density $\rho = 1000\ \text{kg/m}^3$. Point B lies vertically below point A, and the pressure difference between them is 500 Pa.

Ignore atmospheric pressure and take $g = 10\ \text{m/s}^2$.

(a) Which point has the higher hydrostatic pressure?

(b) Find the vertical separation between A and B.

(c) If A is 0.10 m below the water surface, find the gauge pressure at B.

Thus, use force divided by area.

PROBLEM 57

LEVEL III

Adding Water over Mercury to Double Base Pressure

A tall cylinder of cross-sectional area 12.0 cm^2 contains mercury of density 13.6 g/cm^3, whose depth is 8.00 cm.

Water of density 1.00 g/cm^3 is poured on top without mixing. What volume of water must be added to double the gauge pressure at the bottom? Ignore atmospheric pressure.

PROBLEM 58

LEVEL III

Critical Water Height in a Sealed U-Tube

A plastic U-tube has a cross-sectional area of 5.00 cm^2. The short arm, of length $d = 0.800$ m, is filled with water and sealed at the top.

More water is gradually poured into the open long arm. The seal pops off when the net pressure force on it exceeds 9.80 N.

Find the total water-column height in the long arm when the seal is just about to pop.

Take

$$\rho_w = 1000\ \text{kg/m}^3, \qquad g = 9.8\ \text{m/s}^2.$$

4.3 Buoyancy and Floating Bodies

Topic Note: Archimedes' Principle
A body wholly or partially submerged in a fluid experiences a buoyant force equal to the weight of the displaced fluid:

$$F_B = \rho_{\text{fluid}} V_{\text{submerged}} g.$$

For a floating body in equilibrium,

$$\rho_{\text{fluid}} V_{\text{submerged}} g = \rho_{\text{object}} V_{\text{total}} g.$$

Therefore,

$$\frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}.$$

The submerged fraction of a floating body does not depend on gravitational acceleration.

PROBLEM 59

LEVEL I

Floating Object under Varied Gravity

A box floats in a liquid with one-fourth of its volume submerged. The system is moved to a planet where gravitational acceleration is twice that on Earth.

What fraction of the box's volume is submerged there, assuming the liquid and box properties are unchanged?

PROBLEM 60

LEVEL I

Equilibrium Submerged Fraction

A solid object floats in water with exactly half of its total volume submerged.

Find the ratio of the object's density to the density of water.

PROBLEM 61

LEVEL I

Emergent Height of a Floating Cylinder

A uniform solid cylinder has height 10 cm, base area 20 cm^2, and density 0.3 g/cm^3.

It floats vertically in water of density 1.0 g/cm^3.

Find the height of the cylinder above the water surface.

PROBLEM 62

LEVEL II

Submerged Ratio in Different Liquids

A uniform rectangular box floats upright first in water and then in oil.

Express the ratio of its submerged height in water, h_w, to its submerged height in oil, h_o, in terms of the densities ρ_w and ρ_o.

PROBLEM 63

LEVEL II

Cavity Volume in a Cast-Metal Object

An iron casting containing cavities weighs 6000 N in air and 4000 N when fully submerged in water.

Find the total volume of the cavities. The density of solid iron is 7.87 g/cm^3.

Take

$$\rho_w = 1000\ kg/m^3, \qquad g = 9.8\ m/s^2.$$

PROBLEM 64

LEVEL II

Density of a Floating Hollow Cylinder

A hollow cylinder has outer radius 8.0 cm and inner radius 5.0 cm.

It floats in a liquid with half of the outer cylindrical volume submerged. If the liquid density is 0.65 g/cm^3, find the density of the cylinder material.

PROBLEM 65

LEVEL II

Log Raft Capacity

Three children, each weighing 356 N, make a raft from logs of diameter 0.30 m and length 1.80 m.

How many logs are required to keep them afloat in fresh water? The density of the logs is 800 kg/m^3.

Take

$$\rho_w = 1000 \text{ kg/m}^3, \qquad g = 9.8 \text{ m/s}^2.$$

PROBLEM 66

LEVEL III

Floating Body at a Two-Fluid Interface

A solid box floats at the interface between fluid A, with density 1200 kg/m^3, below, and fluid B, with density 800 kg/m^3, above.

The box density is 900 kg/m^3.

Find the fraction of the box's volume submerged in fluid A.

PROBLEM 67

LEVEL III

Apparent Weight in Two Fluids

A rock is suspended from a light string. When fully immersed in water of density 1.00 g/cm^3, the string tension is 1.20 N.

When fully immersed in ethanol of density 0.79 g/cm^3, the tension is 1.60 N.

Calculate the true weight of the rock.

CHAPTER 5

Hints

The hints below correspond to the problems throughout both parts of the collection. They are presented together so that students can attempt the problems independently before consulting the relevant guidance.

Hint 1 | Problem 1 – Level I Answer: (b)

The Idea: Many will use the absolute conversion formula ($F = 1.8C + 32$) and choose 50. However, the question asks for the *change* (ΔT), not an absolute temperature. For differences, we ignore the $+32$ offset and simply multiply by 1.8.

Hint 2 | Problem 2 – Level II Answer: (c)

The Idea: Handle the numerator and denominator separately. Numerator: 1 mg $= 10^{-6}$ kg. Denominator: 1 mm^3 $= (10^{-3}$ m$)^3 = 10^{-9}$ m^3. Dividing the two yields $10^{-6}/10^{-9} = 10^3 = 1,000$.

Hint 3 | Problem 3 – Level II Answer: (c)

The Idea: The second unit is the inverse of the first, multiplied by 100. Inverting 20 km/L gives $1/20$ L/km $= 0.05$ L/km. Multiplying both numerator and denominator by 100 yields 5 L$/100$km.

Hint 4 | Problem 4 – Level II Answer: (c)

The Idea: Converting meters to kilometers requires dividing by 1000. Converting seconds to minutes requires dividing by 60, but since the unit is squared and in the denominator, you divide by 60^2 in the denominator, which is mathematically equivalent to multiplying by 3600 in the numerator.

Hint 5 | Problem 5 – Level I Mathematical Solution:

Use the chain-link conversion method:

$$\text{Rate} = 2\frac{\text{oz}}{\text{hr}} \times \left(\frac{29.57\ \text{cm}^3}{1\ \text{oz}}\right) \times \left(\frac{1000\ \text{mm}^3}{1\ \text{cm}^3}\right) \times \left(\frac{1\ \text{hr}}{3600\ \text{s}}\right)$$

$$\text{Rate} = \frac{2 \cdot 29.57 \cdot 1000}{3600} = \frac{59140}{3600} \approx 16.43\ \text{mm}^3/\text{s}$$

Hint 6 | Problem 6 – Level II Mathematical Solution:

First, express the SI base units in terms of alien units:

$$1\ \text{kg} = \frac{1}{1.5}\ \text{Z} = \frac{2}{3}\ \text{Z}$$

$$1\ \text{m} = \frac{1}{0.5}\ \text{B} = 2\ \text{B}$$

$$1\ \text{s} = \frac{1}{2}\ \text{G} = 0.5\ \text{G}$$

Now substitute these into the Force equation:

$$F = 10\frac{\text{kg}\cdot\text{m}}{\text{s}^2} = 10\frac{(\frac{2}{3}\ \text{Z})\cdot(2\ \text{B})}{(0.5\ \text{G})^2}$$

$$F = 10\frac{4/3}{0.25} = 10\cdot\frac{4}{3}\cdot 4 = 10\cdot\frac{16}{3} = \frac{160}{3} \approx 53.33\ \text{Z}\cdot\text{B}/\text{G}^2$$

Hint 7 | Problem 7 – Level II Mathematical Solution:

Find the conversion factors:

$$1\ \text{Pa} = \frac{1}{101325}\ \text{atm}$$

$$1\ \text{m}^3 = 10^3\ \text{L} \implies 1\ \text{m}^6 = (10^3)^2\ \text{L}^2 = 10^6\ \text{L}^2$$

Substitute into the constant a:

$$a = 0.138\cdot\left(\frac{1}{101325}\right)\cdot(10^6)$$

$$a = \frac{138000}{101325} \approx 1.36\ \text{atm}\cdot\text{L}^2/\text{mol}^2$$

Hint 8 | Problem 8 – Level II Mathematical Solution:

First, convert the radius to parsecs:

$$r = \frac{46.5\times10^9\ \text{ly}}{3.26\ \text{ly/pc}} \approx 14.26\times10^9\ \text{pc}$$

Substitute into the volume formula of a sphere:

$$V = \frac{4}{3}\pi(14.26\times10^9)^3$$

$$V \approx 4.19\cdot(2900\times10^{27}) \approx 1.21\times10^{31}\ \text{pc}^3$$

Hint 9 | Problem 9 – Angle from Vertical Axis If θ is measured from the positive y-axis, then $\tan(\theta) = |v_x/v_y| = 3$. Construct a right triangle or use $\sin(\theta) = \frac{\tan(\theta)}{\sqrt{1+\tan^2(\theta)}}$.

Hint 10 | Problem 10 – Polar Components and Sketch The negative y-axis corresponds to an angle of $270°$. Moving $20°$ clockwise places the vector in the third quadrant at an angle of $250°$ from the positive x-axis.

Hint 11 | Problem 11 – Position Vector Orientation The magnitude is $|\vec{d}| = \sqrt{d_x^2 + d_y^2}$. For the direction, note that $d_x < 0$ and $d_y > 0$, placing the vector in Quadrant II.

Hint 12 | Problem 12 – Force Vector Decomposition Use $F_x = F\cos(\theta)$ and $F_y = F\sin(\theta)$ with $\theta = 120°$.

Hint 13 | Problem 13 – Unit Vector Notation of Force Because the force lies in the xy-plane, $F_z = 0$. Use $F_x = F\cos(45°)$ and $F_y = F\sin(45°)$.

Hint 14 | Problem 14 – 3D Velocity and Direction Angles Direction cosines satisfy $\cos(\alpha) = v_x/v$, $\cos(\beta) = v_y/v$, and $\cos(\gamma) = v_z/v$.

Hint 15 | Problem 15 – Acceleration Vector from Component Ratios Express a_y and a_z in terms of a_x: $a_y = 3.5a_x$ and $a_z = 1.75a_x$. Substitute these into $a^2 = a_x^2 + a_y^2 + a_z^2 = 10^2$ to solve for a_x.

Hint 16 | Problem 16 – Jerk Vector Representation For part (a), if acceleration is constant, its derivative with respect to time is zero. For part (c), recall that the angle with the xz-plane satisfies $\sin(\theta) = |j_y|/j$.

Hint 17 | Problem 17 – Resultant of Two Vectors Add the components: $\vec{R} = (3-3)\hat{i} + \sqrt{55}\hat{j} = \sqrt{55}\hat{j}$. The vector lies purely along the positive y-axis.

Hint 18 | Problem 18 – Vector Subtraction in 3D Rearrange to $\vec{B} = \vec{C} - \vec{A} = (5-0)\hat{i} + (0-2)\hat{j} + (3-4)\hat{k}$.

Hint 19 | Problem 19 – Angle and Resultant Variation Use $C = \sqrt{A^2 + B^2 + 2AB\cos(\theta)}$ and solve for $\cos(\theta)$ in part (a).

Hint 20 | **Problem 20 – Algebraic Vector System and Projections** Equate corresponding components of $\vec{A} - \vec{B}$ with $\vec{C}$. For the x-component: $x - (-2x) = 9 + x \implies 2x = 9$.

Hint 21 | **Problem 21 – Three-Variable Vector Subtraction System** Equate corresponding components of $\vec{A} - \vec{B} - \vec{C} = \vec{D}$. For the x-component: $(2a - a) - 3 = 5 \implies a = 8$.

Hint 22 | **Problem 22 – Angle Between Two Vectors** Compute $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$, then use $\cos(\theta) = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}$.

Hint 23 | **Problem 23 – Self-Dot Product Properties** Use the geometric definition $\vec{V} \cdot \vec{V} = |\vec{V}||\vec{V}|\cos(\theta)$ with $\theta = 0°$, or sum the component products $V_x^2 + V_y^2 + V_z^2$.

Hint 24 | **Problem 24 – Orthogonality Condition** Two non-zero vectors are perpendicular if and only if their dot product equals zero ($\vec{v}_1 \cdot \vec{v}_2 = 0$).

Hint 25 | **Problem 25 – Work Done by a Constant Force** Compute $W = F_x d_x + F_y d_y + F_z d_z$ and use $W = Fd\cos(\theta)$ to find θ.

Hint 26 | **Problem 26 – Vector Projection** The scalar projection is $\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|}$ and the vector projection is $\left(\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|^2}\right)\vec{q}$.

Hint 27 | **Problem 27 – Cross Product Calculation** Expand the 3×3 determinant along the first row.

Hint 28 | **Problem 28 – Self-Cross Product Properties** Recall that the angle between any vector and itself is $\theta = 0°$. Evaluate $|\vec{V} \times \vec{V}| = |\vec{V}||\vec{V}|\sin(0°)$.

Hint 29 | **Problem 29 – Torque** Recall that:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Hint 30 | **Problem 30 – Unit Normal Vector to a Plane** Compute $\vec{n} = \vec{u} \times \vec{v}$ and then normalize it: $\hat{n} = \frac{\vec{n}}{|\vec{n}|}$.

Hint 31 | Problem 31 – Acceleration along a Straight Line Use the kinematic equation relating velocity, acceleration, and displacement:

$$v^2 = v_0^2 + 2a\Delta x.$$

Solve for a.

Hint 32 | Problem 32 – Multi-Stage Average Velocity Average speed is total distance divided by total time:

$$v_{\text{avg}} = \frac{\Delta x_{\text{total}}}{\Delta t_{\text{total}}}.$$

Calculate the time spent in each stage using

$$\Delta t_i = \frac{\Delta x_i}{v_i}.$$

Hint 33 | Problem 33 – Average Velocity over Split Distance and Time First find the effective average velocity v' for the second half of the journey. Since the time intervals are equal,

$$v' = \frac{v_1 + v_2}{2}.$$

Then combine the two equal-distance halves using the harmonic mean:

$$v_{\text{avg}} = \frac{2v_0 v'}{v_0 + v'} = \frac{2v_0(v_1 + v_2)}{2v_0 + v_1 + v_2}.$$

Hint 34 | Problem 34 – Motorboat and Floating Raft Switch to the reference frame of the river water, in which the raft is stationary. The boat moves away from the raft for a time τ and takes the same time τ to return.
The total time elapsed is 2τ, during which the raft floats a distance l. Therefore,

$$v_{\text{flow}} = \frac{l}{2\tau}.$$

Hint 35 | Problem 35 – 2D Motion under Constant Acceleration Express the position vector as a function of time:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2.$$

Separate it into $x(t)$ and $y(t)$ components. Crossing the x-axis occurs when $y(t) = 0$, and crossing the y-axis occurs when $x(t) = 0$. Solve for $t > 0$.

Hint 36 | Problem 36 – 3D Box Corner-to-Corner Fly Displacement The displacement vector is

$$\Delta\vec{r} = L\hat{i} + W\hat{j} + H\hat{k}.$$

Its magnitude follows from the three-dimensional Pythagorean theorem:

$$|\Delta\vec{r}| = \sqrt{L^2 + W^2 + H^2}.$$

Hint 37 | **Problem 37 – Position from Velocity–Time Graph** The displacement equals the area under the velocity–time curve:

$$\Delta x = x(t_2) - x(t_1).$$

Identify the time when $v = 0$, compute the triangular area from $t = 0$ to that instant, and use

$$x(3) - x(0) = \text{Area}.$$

Hint 38 | **Problem 38 – Acceleration from Velocity–Time Graph** Instantaneous acceleration is the slope of the velocity–time graph:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t}.$$

Find the slope of the line segment containing $t = 0$.

Hint 39 | **Problem 39 – Instantaneous Velocity from Position–Time Graph** Instantaneous velocity is the slope of the position–time graph:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}.$$

Calculate the slope of the line segment at $t = 0$.

Hint 40 | **Problem 40 – Continuous Drop of Falling Balls** Let T be the time interval between consecutive drops. The ball striking the ground has fallen for a total time $t_1 = 2T$:

$$H = \frac{1}{2}g(2T)^2 = 2gT^2.$$

The ball currently in the air has fallen for time $t_2 = T$:

$$y_2 = \frac{1}{2}gT^2.$$

Express y_2 as a fraction of H, then calculate its height above the ground:

$$h = H - y_2.$$

Hint 41 | **Problem 41 – Separation of Oppositely Thrown Stones** Choose a common

coordinate system with the origin at the cliff edge and upward positive:

$$y_A(t) = v_0 t - \frac{1}{2} g t^2, \qquad y_B(t) = -v_0 t - \frac{1}{2} g t^2.$$

Subtract the positions to find the separation. Notice that gravity cancels from the relative displacement.

Hint 42 | Problem 42 – Collision of Two Counter-Moving Balls For part (a), use

$$y_1(t) = v_1 t - \frac{1}{2} g t^2, \qquad y_2(t) = H - v_2 t - \frac{1}{2} g t^2.$$

Equate $y_1(t_{\text{coll}}) = y_2(t_{\text{coll}})$ to find the collision time.
For part (b), calculate the time for Ball 1 to reach its apex:

$$t_{\text{apex}} = \frac{v_1}{g}.$$

Compare it with t_{coll}.

Hint 43 | Problem 43 – Diverging Projectiles Distance Write the relative velocity vector between the particles:

$$\vec{v}_{\text{rel}} = \vec{v}_2 - \vec{v}_1.$$

Both particles experience the same gravitational acceleration, so their relative acceleration is zero:

$$\vec{a}_{\text{rel}} = 0.$$

Their relative motion is uniform, giving

$$d(t) = |\vec{v}_{\text{rel}}| t.$$

Hint 44 | Problem 44 – Direct Aim at a Vertical Wall The intended target height is

$$y_{\text{target}} = d \tan\theta.$$

The time to reach the wall is

$$t = \frac{d}{v_0 \cos\theta}.$$

The actual impact height is

$$y_{\text{impact}} = (v_0 \sin\theta) t - \frac{1}{2} g t^2.$$

The vertical deflection is

$$\Delta y = y_{\text{target}} - y_{\text{impact}} = \frac{1}{2} g t^2.$$

Hint 45 | Problem 45 – Cliff-Top Projectiles at Opposite Angles Ball 1 rises, reaches its peak, and returns to the cliff height moving downward at speed v_0 at an angle of 30° below the horizontal.
At that instant, its velocity matches the initial launch velocity of Ball 2. Therefore, the difference in flight times is exactly the time Ball 1 spends executing level-ground projectile motion above the cliff top.

Hint 46 | Problem 46 – Maximum Height to Range Ratio Use

$$H = \frac{v_0^2 \sin^2\theta}{2g}, \qquad R = \frac{v_0^2 \sin(2\theta)}{g} = \frac{2v_0^2 \sin\theta\cos\theta}{g}.$$

Dividing gives

$$\frac{H}{R} = \frac{1}{4}\tan\theta.$$

For part (b), set $H/R = 1/2$.

Hint 47 | Problem 47 – Fixed Initial Speed with Variable Launch Angle The maximum range occurs at 45°:

$$L = \frac{v_0^2}{g}.$$

At 30°,

$$R(30^\circ) = \frac{v_0^2 \sin 60^\circ}{g}.$$

Substitute the expression for L.

Hint 48 | Problem 48 – Level-Ground Projectile with Range 60 m Use

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

to find v_0. The flight time is

$$T = \frac{2v_0 \sin\theta}{g}.$$

For part (c), when $\theta > 45^\circ$, $\sin(2\theta) < 1$, so a higher initial speed is required to achieve the same range. This also increases the flight time.

Hint 49 | Problem 49 – Impact Velocity and Vector Direction On level ground, the horizontal velocity component remains constant, while the vertical component reverses direction:

$$\vec{v}_{fx} = \vec{v}_{ix} = v_i \cos\theta_i \,\hat{i},$$

$$\vec{v}_{fy} = -v_i \sin\theta_i \,\hat{j}.$$

Thus,

$$\vec{v}_i = v_i \cos\theta_i \,\hat{i} + v_i \sin\theta_i \,\hat{j},$$
$$\vec{v}_f = v_i \cos\theta_i \,\hat{i} - v_i \sin\theta_i \,\hat{j}.$$

Hint 50 | Problem 50 – Vertical Component of Impact Velocity against a Wall From the graph, $v_{y0} = v_{ys} = 5.0\ \mathrm{m/s}$. The vertical velocity reaches zero at $x = 10\ \mathrm{m}$, the horizontal location of the trajectory's peak.
Use

$$v_y(t) = v_{y0} - gt = 0$$

to find the time to the peak:

$$t_p = \frac{v_{y0}}{g}.$$

Then use $x(t_p) = v_x t_p = 10\ \mathrm{m}$ to find v_x. Finally,

$$\tan\theta_0 = \frac{v_{y0}}{v_x}.$$

Hint 51 | Problem 51 – Density Unit Conversion I Use

$$1\ \mathrm{g/cm^3} = 1000\ \mathrm{kg/m^3}.$$

Divide the given value by 1000.

Hint 52 | Problem 52 – Density Unit Conversion II Divide the density of water by the density represented by one R.

Hint 53 | Problem 53 – Pressure Under a Heavy Metallic Cylinder Use

$$P = \frac{F}{A} = \frac{mg}{A}.$$

Convert the area using

$$1\ \mathrm{cm^2} = 10^{-4}\ \mathrm{m^2}.$$

Hint 54 | Problem 54 – Force Scaling in a Hydraulic Press Apply Pascal's principle:

$$\frac{f}{a} = \frac{F}{A}.$$

Solve for F.

Hint 55 | Problem 55 – Circular Hydraulic Lift The ratio of circular areas is the ratio of

squared diameters:

$$\frac{A_2}{A_1} = \left(\frac{d_2}{d_1}\right)^2.$$

Use Pascal's principle.

Hint 56 | Problem 56 – Hydrostatic Pressure Difference and Depth Pressure increases with depth:

$$\Delta P = \rho g \Delta h.$$

The gauge pressure at A is $\rho g h_A$. Add the pressure difference to obtain the pressure at B.

Hint 57 | Problem 57 – Adding Water over Mercury to Double Base Pressure Initially,

$$P_i = \rho_m g h_m.$$

The added water must contribute an equal pressure:

$$\rho_w g h_w = \rho_m g h_m.$$

Then use $V_w = A h_w$, keeping units consistent.

Hint 58 | Problem 58 – Critical Water Height in a Sealed U-Tube The maximum gauge pressure difference is

$$P_{\max} = \frac{F_{\max}}{A}.$$

It equals $\rho_w g \Delta h$, where Δh is the difference between the two water levels. The total height in the long arm is

$$H = d + \Delta h.$$

Hint 59 | Problem 59 – Floating Object under Varied Gravity Set the buoyant force equal to the object's weight. The factor g cancels.

Hint 60 | Problem 60 – Equilibrium Submerged Fraction Equate the object's weight to the buoyant force and solve for ρ_{obj}/ρ_w.

Hint 61 | Problem 61 – Emergent Height of a Floating Cylinder The submerged height is

$$h_{\text{sub}} = \frac{\rho_{\text{cyl}}}{\rho_w} h_{\text{total}}.$$

Subtract it from the total height.

Hint 62 | **Problem 62 – Submerged Ratio in Different Liquids** The box's weight is unchanged. Equate the buoyant forces:

$$\rho_w h_w A g = \rho_o h_o A g.$$

Hint 63 | **Problem 63 – Cavity Volume in a Cast-Metal Object** The buoyant force is

$$F_B = W_{\text{air}} - W_{\text{water}},$$

which gives the casting's external volume.
The iron volume is

$$V_{\text{iron}} = \frac{W_{\text{air}}}{\rho_{\text{iron}} g}.$$

Subtract the iron volume from the external volume to find the cavity volume. Convert the iron density to SI units first.

Hint 64 | **Problem 64 – Density of a Floating Hollow Cylinder** Equate the buoyant force,

$$\rho_\ell g \left(\frac{1}{2}\pi R^2 L\right),$$

to the cylinder's weight,

$$\rho_m g \pi (R^2 - r^2) L.$$

Cancel common factors and solve for ρ_m.

Hint 65 | **Problem 65 – Log Raft Capacity** For N logs, the maximum buoyancy when the logs are fully submerged must be at least the weight of the logs plus the children:

$$N \rho_w V g \geq N \rho_{\text{log}} V g + 3 W_c,$$

where

$$V = \pi (d/2)^2 L.$$

Solve for the smallest integer N.

Hint 66 | **Problem 66 – Floating Body at a Two-Fluid Interface** Let f be the fraction in fluid A; the fraction in fluid B is $1 - f$.
Use

$$f \rho_A + (1 - f) \rho_B = \rho_{\text{box}}$$

and solve for f.

Hint 67 | **Problem 67 – Apparent Weight in Two Fluids** Write

$$T_w = W - \rho_w V g, \qquad T_e = W - \rho_e V g.$$

Subtract the equations to find Vg, then substitute into either equation to obtain W. Keep density units consistent.